Structure and Function of the Renal System and Urinary Tract

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Renal and urinary tract structure → Urine formation (filtration, reabsorption, secretion) → GFR → Fluid, electrolyte, and acid-base regulation → Blood pressure → Endocrine functions → Compensatory mechanisms

This lecture reviews normal kidney and urinary tract anatomy and physiology from your prerequisites. It sets the stage for the disorder lectures that follow. Tie it to what you already know about fluid and electrolytes, blood pressure, and acid-base balance, because those concepts come together in the kidneys.

Three things to keep in mind: (1) Blood has to reach the kidneys. (2) The nephrons have to filter and process that blood. (3) Urine needs an open pathway out of the body. A problem at any of these points can affect the whole system, and knowing where it starts helps you explain the symptoms and the lab findings.

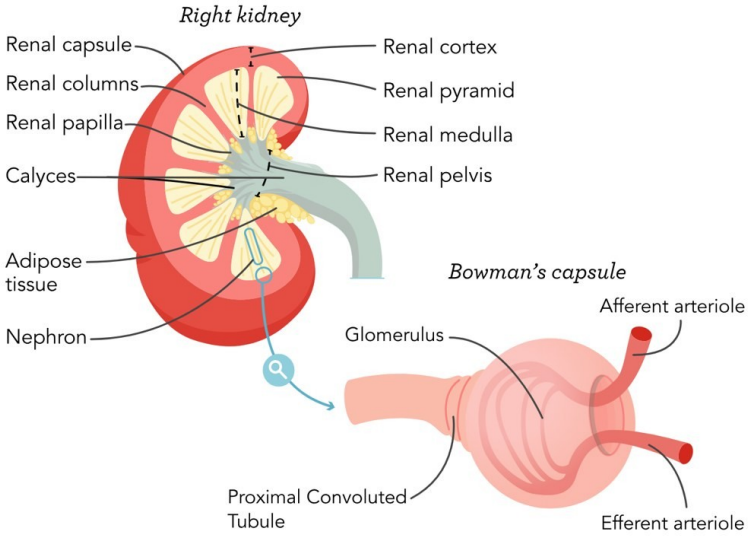
Learning Objectives

- Discuss the basic structure of the renal system and urinary tract.
- Examine the physiology of urine formation.
- Explore the regulation of fluid and electrolyte balance by the renal system.
- Correlate the renal system's role in blood pressure regulation.
- Discuss the endocrine functions of the renal system.
- Relate the body's compensatory mechanisms in response to renal system challenges.

A note on terminology: This module uses "male" and "female" to describe anatomy because that is the language of the ATI Engage modules and other healthcare resources. In this course the terms refer to anatomical structures and physiology, not gender identity. Sex and gender are distinct, and people have diverse gender identities that binary language does not fully represent. Use anatomy-specific and person-centered language in clinical practice.

Structure of the Renal System and Urinary Tract

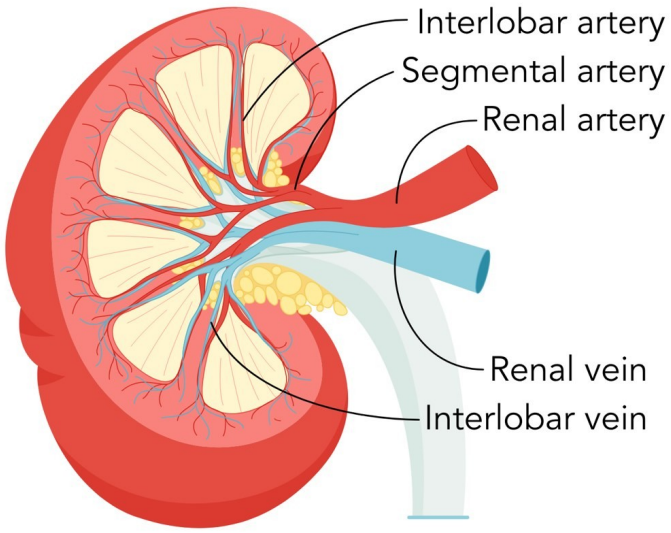
The renal system includes the kidneys, ureters, bladder, and urethra. The kidneys filter the blood and produce urine.



Kidney structure and the nephron/Bowman's capsule (lecture slide)

Kidneys and Blood Flow

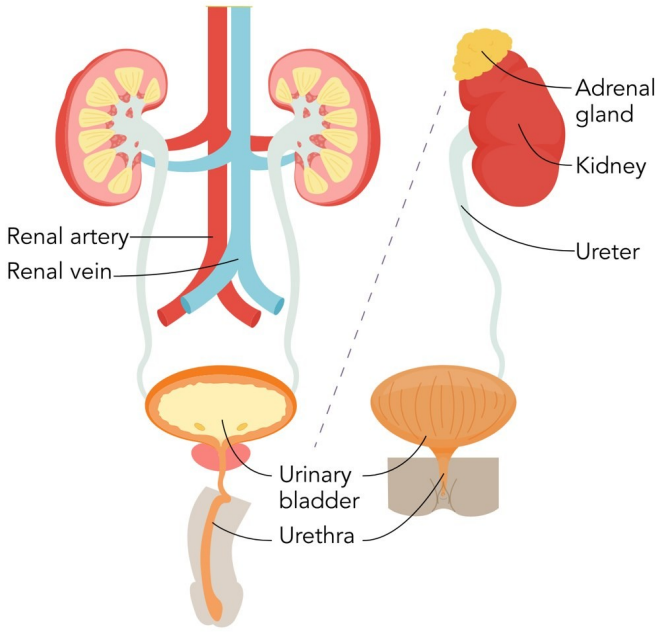
1. **Arrival:** blood travels from the aorta through the renal arteries and enters each kidney at the hilum.
2. **Branching:** inside the kidney the arteries branch into smaller vessels that deliver blood to the nephrons, the kidney's functional units.
3. **Filtration:** in the glomerulus (a cluster of capillaries surrounded by Bowman's capsule), water and small substances filter from the blood into the capsule.
4. **Return:** the blood continues through the circulation and leaves the kidney through the renal vein, back into circulation.
5. **Processing:** the filtered fluid moves through the renal tubules, where the body takes back what it needs and adds wastes it needs to get rid of. The fluid that remains becomes urine.



Renal artery, interlobar and segmental arteries, renal vein (lecture slide)

Path of Urine

Collecting ducts → calyces → renal pelvis → ureters → bladder (storage) → urethra → out of the body (assuming typical anatomy).



Urinary tract: kidneys, ureters, bladder, urethra (lecture slide)

- **Ureters:** muscular walls contract in waves (peristalsis) to move urine from each kidney toward the bladder.
- **Bladder:** stores urine until we urinate.
- **Urethra:** carries urine from the bladder to the outside of the body.

Know the difference: Ureters carry urine from the kidneys to the bladder. The urethra carries urine from the bladder to outside the body. When you study urinary disorders, the location of an infection or obstruction helps explain the symptoms.

Discussion Question 1: Why are UTIs more common in people with a vagina?

Factor	Why it raises UTI risk
Shorter urethra	About 4 cm (about 1.5 inches) with typical vulvar anatomy, versus about 20 cm (8 inches) in people with a penis. Bacteria have less distance to travel to reach the bladder
Location of the urethral opening	Close to the anus, where intestinal bacteria can become a source of infection. This is a very common source of infection

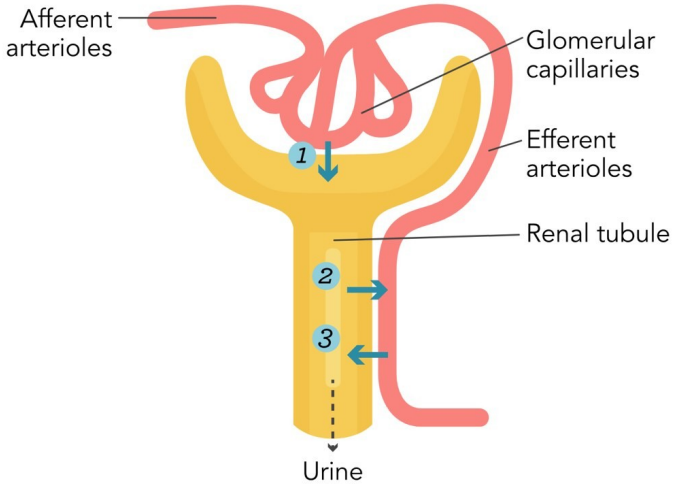
This helps explain why lower urinary tract infections are more common with this anatomy. The same ascending pathway returns when cystitis and pyelonephritis are covered.

Physiology of Urine Formation

Urine formation involves three processes within the kidney. The key idea is the direction substances move.

Process	Direction	Purpose
Filtration	Blood → Bowman's capsule	Pulls water and small substances out of the blood
Reabsorption	Tubular fluid → blood	Recovers water and useful substances (things we need to keep). Happens continuously
Secretion	Blood → tubular fluid	Moves selected substances into the fluid so they leave in the urine

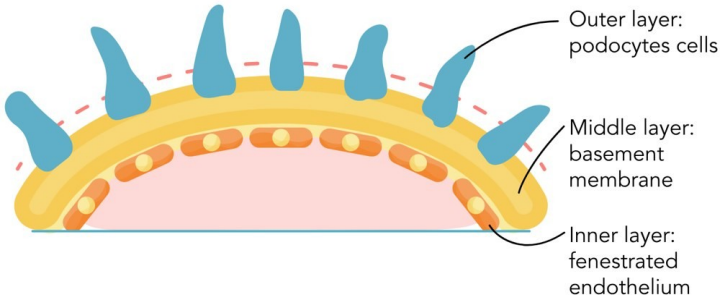
Together these decide what the body keeps and what it excretes, continuously, as fluid moves through the nephron.



Nephron: filtration, reabsorption, secretion, and urine (lecture slide)

Filtration

- **Blood flow:** blood enters the glomerulus through the afferent arteriole and leaves through the efferent arteriole. Memory aid: A before E, so afferent comes in and efferent exits.



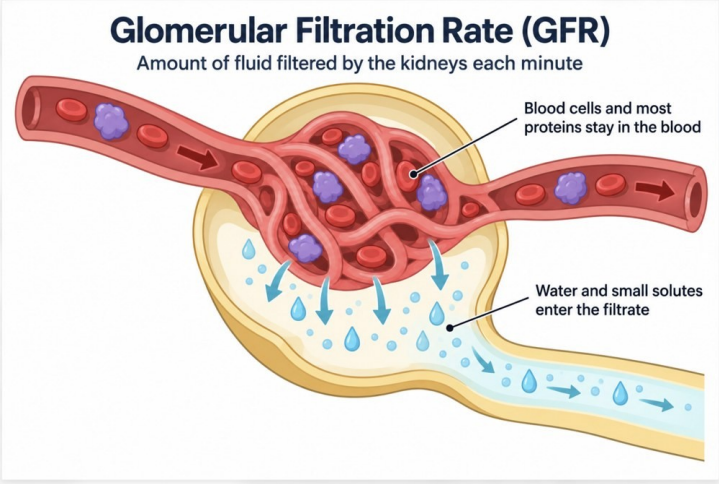
Filtration barrier: endothelium, basement membrane, podocytes (lecture slide)

- **Filtration barrier:** capillary endothelium (fenestrated inner layer), basement membrane (middle layer), and podocyte foot processes (outer layer).

Key point: A normal barrier lets water and small solutes pass while keeping blood cells and most proteins in the circulation. So normal means no blood and no protein in the urine. The instructor repeated this because it becomes important in the kidney disorders.

- **Pressures:** hydrostatic pressure in the glomerular capillaries pushes fluid outward. Plasma proteins create oncotic pressure that opposes that movement by holding water in the bloodstream (from the fluid balance content).

Glomerular Filtration Rate (GFR)



GFR: amount of fluid filtered by the kidneys each minute (lecture slide)

- **Definition:** how much fluid the glomeruli filter each minute. It indicates kidney filtering function and kidney function in general.
- **Requirements:** adequate blood flow and an intact filtration barrier.
- **When GFR falls:** less fluid is filtered and waste products can accumulate in the blood.
- **eGFR:** in practice we use the estimated GFR, calculated from blood markers such as creatinine. This is what appears in lab values.

Clinical question to ask: When you study kidney disorders, ask whether the problem involves blood reaching the filter, damage to the filter, or something happening further along the pathway.

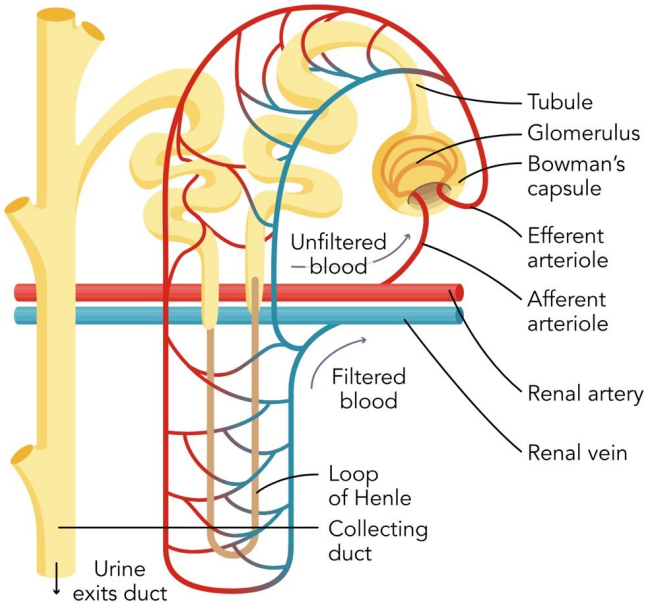
Note on GFR: Removal of Race from the eGFR Equation

- **Past practice:** older equations included a correction factor for Black patients only. It automatically raised the reported GFR, making kidney function look better than it was.
- **Why it was wrong:** it assumed Black patients have higher creatinine because of greater muscle mass. That assumption was not based on evidence. Race is a social construct, not a biological category.
- **Harm:** an artificially higher GFR could delay diagnosis of chronic kidney disease, referral to a specialist, and listing for a transplant.
- **Change:** in 2021 a joint task force of the National Kidney Foundation and the American Society of Nephrology recommended removing race from the calculation. Much of the push came from a Black woman medical student who asked why only Black patients had an adjusted GFR and got faculty buy-in, followed by advocacy at higher levels.

Takeaway: Every adult gets one eGFR regardless of race. Where more precision is needed, a second marker can be added for the individual, not based on race. This gives more accurate and equitable kidney care.

Reabsorption and Secretion in the Tubules

Filtration pulls many useful substances into the tubular fluid along with waste. The system is not perfect, so the tubules adjust the filtrate's composition and put needed substances back into the body.



Nephron with loop of Henle, collecting duct, and blood supply (lecture slide)

Segment	What happens
Proximal tubule	Reabsorbs much of the filtered water and sodium. Under normal conditions it also recovers nearly all filtered glucose and amino acids
Loop of Henle: descending limb	Water can leave. This concentrates the urine and protects the body from dehydration
Loop of Henle: ascending limb	Reabsorbs salts and lets very little water flow. This builds the concentration gradient that lets the kidneys conserve water
Distal tubule and collecting ducts	Further adjustments and refinements, including responses to hormones
Secretion	Moves hydrogen ions, potassium, and certain medications from the blood into the tubular fluid, to dispose of substances that are not useful or are harmful (the slide mentions urea and uric acid). Connects to pharmacology and how medications are excreted

Remember the direction: Reabsorption returns substances to the blood. Secretion adds substances to the fluid that will become urine.

Discussion Question 2: How do reabsorption and secretion maintain fluid balance?

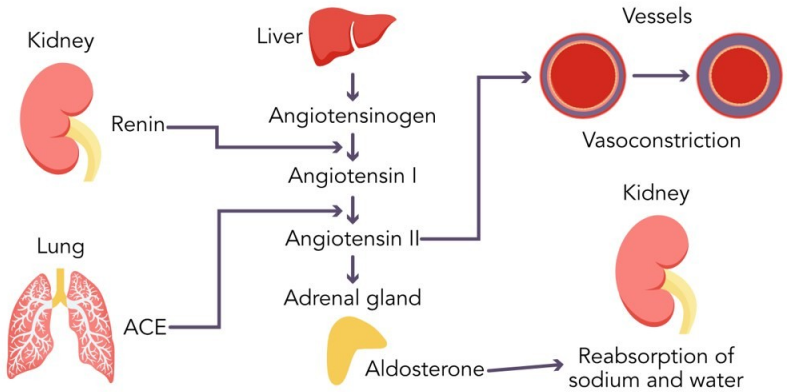
- **Why we do not lose it all:** the kidneys filter water and electrolytes, but reabsorption returns needed water and substances to the circulation. Movement in both directions maintains balance.
- **Secretion:** adjusts removal of selected substances, including potassium, hydrogen ions, and medications.
- **Fluid volume:** adjusting sodium and water reabsorption is especially important. Fluid volume does not come from the stomach; the kidneys adjust it to keep you homeostatic. This is the big connection back to the cardiac lectures.
- **If the tubules are damaged:** even with some filtration still occurring, these adjustments are disrupted and urine production is not normal. Urine production alone does not show that kidney function is normal.

Regulation of Fluid, Electrolyte, and Acid-Base Balance

ADH

1. **Trigger:** blood becomes more concentrated, or circulating volume falls substantially.
2. **Release:** the posterior pituitary releases ADH.
3. **Effect:** ADH makes the collecting ducts more permeable to water, so more water returns to the bloodstream.
4. **Result:** urine volume decreases and urine becomes more concentrated.

RAAS



Renin-angiotensin-aldosterone system (lecture slide)

1. **Trigger:** reduced kidney perfusion triggers renin release.
2. **Angiotensin I:** renin begins a sequence that produces angiotensin I.
3. **Angiotensin II:** ACE converts angiotensin I to angiotensin II.
4. **Effects:** angiotensin II constricts blood vessels and stimulates aldosterone release from the adrenal cortex. Aldosterone promotes sodium reabsorption and potassium secretion in the later parts of the nephron.
5. **Water:** sodium retention supports water retention, because water follows sodium.

Exam note: ADH conserves water. Aldosterone promotes sodium retention and potassium excretion.

Discussion Question 3: How does hydration status affect urine?

	ADH	Urine
Dehydration (more concentrated blood, lower volume)	Increases, so the kidneys keep more water	Smaller volume, more concentrated
Excess water intake	Decreases, so less water returns to blood in the collecting ducts	Larger volume, more dilute

This is a normal compensatory response. Later, consider what happens when kidney damage limits the ability to do this.

Minerals

- **Adjust reabsorption and excretion:** the kidneys regulate calcium, phosphate, and magnesium, which support nerve signaling, muscle activity, and bone health.
- **Hormones and vitamin D:** calcium and phosphate regulation also involve hormones and vitamin D. Keep this in mind for chronic kidney disease.

Acid-Base Equilibrium

- **Normal blood pH:** 7.35 to 7.45.
- **Bicarbonate:** the kidneys recover (reabsorb) filtered bicarbonate, which acts as a buffer.
- **Acid excretion:** excreting acid allows the kidney to add new bicarbonate to the blood and adjust pH.

Organ	Acid-base role	Speed
Lungs	Regulate carbon dioxide	Fast (blow off CO2 quickly)
Kidneys	Regulate bicarbonate and acid excretion: more bicarbonate reabsorption, more acid excretion	Slower: takes much longer to respond

Discussion Question 4: How does the renal system maintain acid-base balance?

- **If blood is too acidic:** functioning kidneys increase acid (hydrogen ion) excretion, conserve bicarbonate, and generate new bicarbonate during acid excretion.
- **If kidney function is impaired:** the body struggles to eliminate the acid load. Acid accumulates and bicarbonate falls, which can lead to or contribute to metabolic acidosis.

Key connection: Kidney dysfunction affects the chemical balance of the blood as well as urine production.

Role in Blood Pressure Regulation



DASH diet pattern: reduces sodium intake to support blood pressure control (lecture slide)

- **Sodium and water:** the kidneys influence blood pressure over time by regulating sodium and water balance. Retaining sodium and water raises extracellular fluid volume and circulating volume, which can raise blood pressure. Excreting more sodium and water lowers volume.
- **Low perfusion:** when kidney perfusion falls, RAAS helps support pressure and volume.
- **Pressure natriuresis:** when perfusion pressure rises, functioning kidneys generally increase sodium excretion.
- **Diet:** sodium intake determines how much sodium the kidneys must handle (more salt in, more salt to process). The DASH diet is typically recommended for hypertension. The focus here is the underlying physiology, not the dietary behavior.

Vicious cycle: The relationship works in both directions. Kidney dysfunction can contribute to hypertension, and persistent hypertension can damage the kidneys.

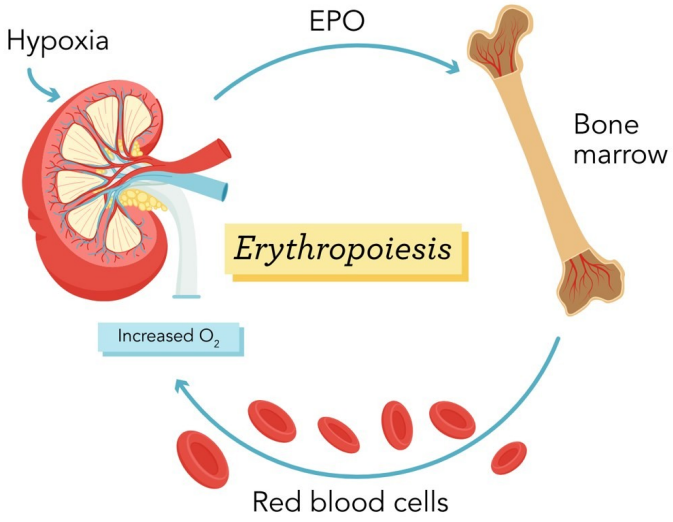
Discussion Question 5: Why does traumatic blood loss cause hypotension and decreased urine output?

1. **Volume loss:** blood loss reduces circulating volume and venous return.
2. **Pressure drop:** cardiac output and blood pressure fall.
3. **Perfusion drop:** less blood reaches the kidneys, so GFR falls.
4. **Compensation:** the body activates RAAS and ADH to conserve sodium and water. Together these reduce urine output.

Is low urine output proof the kidneys are injured? Not necessarily. Initially it is a problem of blood reaching the kidneys. If poor perfusion continues, the kidneys also lose oxygen delivery and the tissue can become injured. This distinction leads into prerenal acute kidney injury later. The same applies to vasoconstriction in hypertension: less blood flow means less urine output and less oxygen-rich blood reaching kidney tissue.

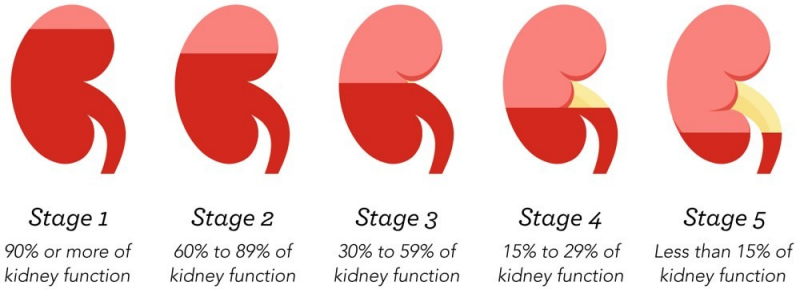
Impact of the Renal System on Endocrine Function

Two examples: vitamin D activation and erythropoietin (EPO) production.



Hypoxia triggers EPO, which stimulates red blood cell production in bone marrow (lecture slide)

Hormone	Normal role	In chronic kidney disease (CKD)
Calcitriol (active vitamin D)	The kidneys complete activation of vitamin D into calcitriol, which supports calcium absorption from the intestines	Less activation, so less calcium absorbed and disturbed calcium balance. Difficulty excreting phosphate adds to the mineral imbalance and affects bone health
Erythropoietin (EPO)	Signals the bone marrow to make red blood cells	Damaged kidneys make less EPO, contributing to anemia. Other factors contribute, but reduced EPO is the main renal connection



CKD stages 1 to 5 by percent of kidney function (lecture slide)

The CKD stages image links declining kidney function to effects throughout the body. Stages and complications are covered in the kidney disorders lecture.

Discussion Question 7: Why do CKD patients develop vitamin D and calcium deficiencies and anemia?

- **Calcium:** reduced calcitriol means less intestinal calcium absorption. Phosphate retention further disrupts mineral balance and contributes to bone disease.
- **Anemia:** reduced EPO means less stimulation of the bone marrow to produce red blood cells.

Loss of kidney function affects other organs and tissues, which is why kidney disease produces findings beyond changes in urine.

Compensatory Mechanisms

- **Water limited:** hormone signals promote water conservation.
- **Excess water:** the kidneys increase excretion.
- **Acid-base change:** the kidneys adjust acid and bicarbonate.

Key point: These responses depend on adequate kidney function. You cannot produce adequate compensatory mechanisms without it. During acute kidney injury (AKI), filtration or tubular function may become impaired, which limits the body's ability to regulate fluid, electrolytes, and acid-base balance.

Keep asking what the body is trying to correct and whether the kidneys can carry out the response. A compensatory signal may be present even when the kidneys cannot respond effectively. Is the problem the signal, or can the kidney not respond to the signal?

Discussion Question 6: How does the body compensate for fluid volume imbalances?

Situation	Normal kidney response	If kidney function is impaired
Fluid loss (vomiting, diarrhea)	Promote sodium and water conservation to support circulating volume	Limited or absent response: difficulty conserving water
Excess water intake	Increase water excretion	Retained excess fluid

Interpret urine output alongside the person's overall fluid status.

Wrapping Up

As you move into the disorders, use today's physiology to explain the findings. Ask where the problem begins:

- Is blood reaching the kidneys?
- Is the filtration barrier intact?
- Can the tubules adjust the filtrate?
- Can urine drain normally?

Then connect the problem to what happens in the body. The goal is to move from remembering a disorder's symptoms to explaining why they occur.

Quick Review

Topic	Key points
Pathway	Aorta → renal artery → hilum → nephron → renal vein. Urine: collecting ducts → calyces → renal pelvis → ureters → bladder → urethra
Ureter vs. urethra	Ureters: kidney to bladder. Urethra: bladder to outside
UTI risk with vagina	Urethra about 4 cm (vs. about 20 cm); opening near the anus
Urine formation	Filtration (blood to capsule), reabsorption (tubule to blood), secretion (blood to tubule)
Filtration barrier	Endothelium, basement membrane, podocytes. Keeps blood cells and most proteins in blood; normal urine has no blood or protein
Afferent / efferent	A before E: afferent in, efferent exits
GFR	Fluid filtered per minute; eGFR from creatinine; one eGFR for all adults (race removed in 2021 recommendation)
Tubules	Proximal: water, sodium, glucose, amino acids. Descending limb: water leaves. Ascending: salts, little water. Distal and collecting ducts: hormone-responsive
ADH	Posterior pituitary; more water permeability in collecting ducts; less urine, more concentrated
RAAS	Low perfusion → renin → angiotensin I → ACE → angiotensin II → vasoconstriction and aldosterone → Na reabsorption, K secretion
Acid-base	Kidneys reabsorb bicarbonate and excrete acid; slower than lungs. Impaired kidneys: metabolic acidosis
Blood pressure	Sodium and water balance; pressure natriuresis; HTN and kidney damage form a vicious cycle
Endocrine	Calcitriol (calcium absorption) and EPO (red blood cells): both reduced in CKD
Compensation	Requires adequate kidney function; limited in AKI and CKD

Notes on Transcript Cleanup

Corrected: "R**S" is written as RAAS. A garbled phrase about the filtration barrier ("filtration sits between the foot process here") was written as the three layers on the slide. Repeated points were combined. Spoken pause-and-discuss prompts are kept as Discussion Question sections with the instructor's answers; Questions 6 and 7 were asked together near the end, and each is placed under its topic here. The slide says the descending limb "concentrates urine," while the instructor said water leaves it; both are included.